

Bidirectional EN ↔ ES translation transformer

ENCODER

DECODER

Reading the diagram

— data flow

- - - residual connection

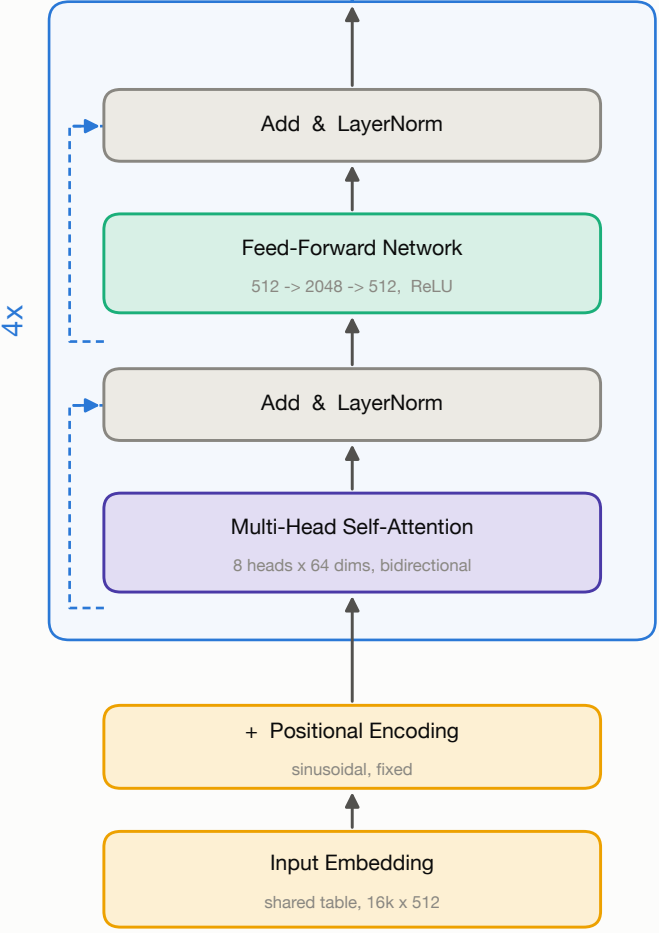
- - - encoder to decoder

37.6M parameters
4 encoder + 4 decoder layers
d_model 512, 8 heads, d_ff 2048
one shared 16k vocabulary
embeddings tied 3 ways

the only path from source to target

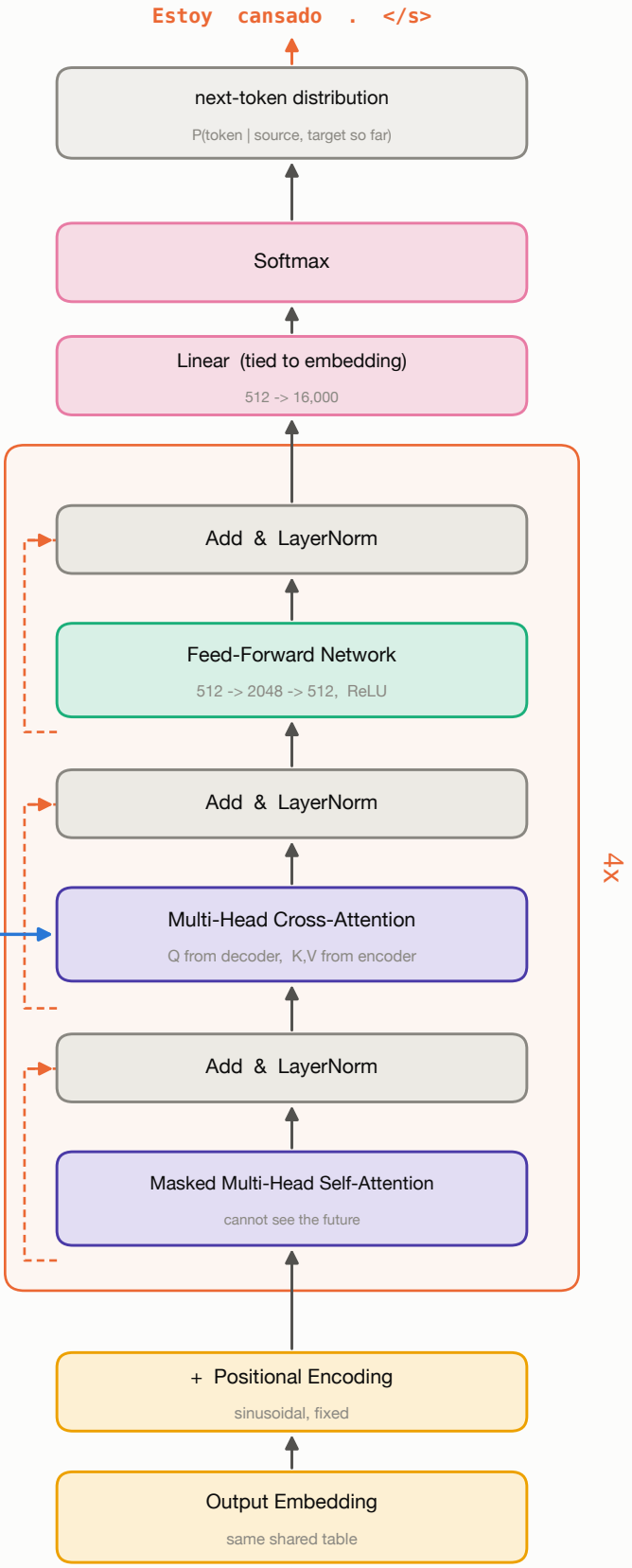
memory (6, 512)

K, V



source: <2es> I am tired . </s>

the direction tag tells one model which way to translate



target so far: <s> Estoy cansado .

shifted right: position t predicts token t+1

Estoy cansado . </s>